

When the US entered World War II in 1941, Oppenheimer took up atomic weapons research, which he judged “a good, honest, practical way” to use science. By July 1942, he had emerged as the natural leader of a group of scientists convened by the Manhattan Project, the Allied nuclear-weapons program.

Atomic pioneer

Oppenheimer’s ambition and intellect were well suited to the vast scale of the project’s interdisciplinary collaboration, and in 1943, he was made director of

Oppenheimer directed the huge-scale, complex Project Y, which grew to have more than 6,000 staff, including the highest concentration of Nobel laureates ever gathered in one place.



VANNEVAR BUSH

Successful inventor and engineer Vannevar Bush was an influential figure in US science administration—in particular, the Manhattan Project—during and after World War II.



Bush (1890–1974) was tasked with leading the National Defense Research Committee by President Roosevelt in 1940. He informed the president of the 1941 UK MAUD Report, which had concluded that an atomic bomb was feasible within the expected timescale of World War II, leading directly to the creation of the Manhattan Project. In 1945, he outlined an idea for a desklike computing device, called a memex, that helped inspire Tim Berners-Lee’s World Wide Web some 40 years later.

Project Y, the secret facility in New Mexico tasked with making the atomic bomb. His leadership and grasp of scientific, military, and engineering detail were integral to the project’s success.

Two types of weapons were perfected: one using the “gun-type” uranium method and the other the “implosion-type” plutonium method of triggering a nuclear chain reaction. The former was used in the bomb that obliterated Hiroshima, Japan, on August 6, 1945, and the latter was used in the Trinity test of July 16 and the bomb dropped on Nagasaki, Japan, on August 9, which precipitated Japan’s surrender.

Peacetime remorse

After initial euphoria, Oppenheimer was horrified at the devastation the bombs caused. He resigned from the Manhattan Project later in 1945, and instead mixed academia and atomic consultancy. After he opposed the development of the H-bomb, opinion turned against him; he lost his security clearance in 1953 and was dismissed from the US Atomic Energy Commission.

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